

Supplementary Material S3

TRIPOD Checklist — Transparent Reporting of a Multivariable Prediction Model

for Individual Prognosis Or Diagnosis (TRIPOD) Statement

Collins et al. (2015) — 22-Item Checklist for Prediction Model Development Studies

Paper Title: Explainable Machine Learning for Student Burnout Classification and Risk Stratification: A Mixed-Methods Study with Qualitative Triangulation

Authors: Rifat Miah (rifatmiah1992003@gmail.com), Dr. A.S.M. Shihavuddin (shihav@eee.green.edu.bd)

Corresponding Author: Rifat Miah (Email: rifatmiah1992003@gmail.com), Department of Computer Science and Engineering, Presidency University, Dhaka 1212, Bangladesh (Institutional Contact: info@pu.edu.bd)

Model Type: Classification (Prediction Model Development — 10-Fold Stratified Cross-Validation)

Outcome: Binary classification of High Academic Burnout (burnout_score = 3) vs. Low/Medium Burnout (burnout_score = 1 or 2) in Bangladeshi university students.

TRIPOD Checklist

Section	Item No.	Checklist Item	Manuscript Location	Reported?
Title and Abstract	1	Identify the study as developing and/or validating a multivariable prediction model, the target population, and the outcome to be predicted.	Title, Abstract (Methods paragraph)	✓Yes
Introduction	2a	Explain the medical context (including whether diagnostic or prognostic) and rationale for developing or validating the multivariable prediction model, including references	Section 1.1, Section 2.4, Table 1	✓Yes

		to existing models.		
	2b	Specify the objectives, including whether the study describes the development or validation of the model or both.	Section 1.7, Section 1.4 (H1–H4)	✔Yes
Methods — Source of Data	3	Describe the study design or source of data (e.g., randomised trial, cohort, or registry data), separately for the development and validation data sets, if applicable.	Section 3.1, Section 3.2	✔Yes
	4	Describe the eligibility criteria for participants.	Section 3.2, Section 3.4 (Ethics; 18+ enrolled students)	✔Yes
	5	Describe the outcome(s) to be predicted.	Section 3.3, Section 3.3.1, Table 2 (burnout_score variable)	✔Yes
	6	Report the number of participants with and without the outcome.	Section 3.5 (High: n = 255, 42.43%; Low/Med: n = 346, 57.57%)	✔Yes
	7	Describe the predictors used in the model, including how and when they were measured.	Section 3.3 (Table 2), Section 3.5 (Table 3, composite indices)	✔Yes
	8	Describe the sample size, including any sample size calculations.	Section 3.2 (N = 601); Appendix / power note in Section 3.2 (G*Power calculation referenced, minimum n = 452 per Cohen, 1988 [118])	✔Yes
	9	Describe how missing data were	Section 3.5 ("no missing values for any of the 18	✔Yes

		handled.	variables")	
Methods — Model Development	10a	Describe how predictors were handled in the analyses.	Section 3.5 (StandardScaler, OneHotEncoder, leak-free pipeline), Section 3.6	✔Yes
	10b	Specify the type of model, all model-building procedures (including any predictor selection), and method for internal validation.	Section 3.6 (10 classifiers + Soft Voting Ensemble described; GridSearchCV nested CV for hyperparameter selection)	✔Yes
	10c	For validation, describe the method of performance assessment.	Section 3.6, Section 6 (10-fold stratified CV, McNemar test, Bonferroni correction; Table 4–Table 6)	✔Yes
Methods — Statistics	11	Describe all measures used to assess model performance and, if relevant, to compare multiple models.	Section 3.6, Section 6 (Accuracy, Balanced Accuracy, ROC-AUC, F1, Precision, Recall, MCC, Brier Score; McNemar pairwise tests reported in Table 5)	✔Yes
Results — Participants	12	Report the number of participants and outcome events in the overall data set and, if applicable, in the development and validation data sets.	Section 4.1, Section 6.1 (Table 4: n = 601; High Burnout n = 255; Low/Med n = 346)	✔Yes
	13	Describe the characteristics of the participants (basic demographics, clinical characteristics) and the distribution of predictors.	Section 4 (full EDA: Sections 4.1–4.4), Table 2, Table 3	✔Yes

Results — Model Development	14a	Specify the number of participants and outcome events in each analysis.	Section 6.1, Table 4	✓Yes
	14b	If done, report the results from any model selection procedure.	Section 3.6 (all 10 models benchmarked; champion model selected by Accuracy+AUC; nested CV hyperparameter tuning described)	✓Yes
	15	Present the full prediction model to allow predictions for individuals (i.e., all regression coefficients, and model intercept or baseline survival at a given time point).	Section 7 (SHAP global feature importance Table 7a, SHAP local attributions Figure 6); the full trained Random Forest model is available at the GitHub repository. <i>Note: For tree-ensemble models, the equivalent of regression coefficients is the full SHAP value decomposition, which is reported in Table 7a and Figure 6.</i>	✓Yes
	16	Report performance measures of the prediction model.	Section 6.1 (Table 4: 10-fold CV metrics for all models); Section 6.2 (confusion matrix analysis, threshold calibration)	✓Yes
Discussion	17	Discuss the limitations of the study (e.g., nonrepresentative sample, few events per variable, missing data).	Section 10 (full limitations section: single-item criterion, cross-sectional design, convenience sampling, geographic concentration,	✓Yes

			pseudo-external validation only, absence of pre-registration)	
Other Information	18	Provide information about funding sources and the role of the funders for the present study.	<i>To be added to final manuscript:</i> This research received no external funding. The study was conducted as part of the academic research activities of the Department of Computer Science and Engineering, Presidency University.	☑ To Add
	19	Provide supplementary information (e.g., study protocol, statistical analysis plan, or patient consent forms).	Supplementary Material S1 (Interview Guide), S2 (COREQ Checklist), S3 (TRIPOD Checklist) — GitHub repository link provided in Section 3.4.	✔ Yes

☑ **Action Required for Item 18:** Add the following sentence to the manuscript (Acknowledgements section or footnote): *"This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors. The study was conducted as part of independent academic research activities."*

Explanatory Notes on Key Items

Item 10b — Model Building Procedure

Ten classification models were evaluated: Logistic Regression, Decision Tree, Random Forest, Extra Trees, Gradient Boosting, XGBoost, LightGBM, CatBoost, Support Vector Machine, and Multi-layer Perceptron, plus a Soft Voting Ensemble. Hyperparameter tuning was performed via nested 10-fold outer / 5-fold inner cross-validation using GridSearchCV on the top models. All pipelines were implemented in scikit-learn with ColumnTransformer preprocessing to prevent data leakage.

Item 15 — Full Prediction Model

For tree-ensemble models, SHAP (SHapley Additive exPlanations) values represent the functionally equivalent construct to regression coefficients, providing both global feature importance (Table 7a) and individual-level attributions (Figure 6). The full trained Random Forest model (serialised in .pk1 format), all pipeline components, and SHAP explainer objects are publicly available at: <https://github.com/rifatmiah92/student-burnout-ml-mixed-methods>

Item 16 — Performance Measures

Primary metrics: Accuracy, Balanced Accuracy, ROC-AUC, Macro-F1, Precision, Recall, Matthews Correlation Coefficient (MCC), Brier Score. All reported as mean $\pm$ SD across 10 folds. Pairwise McNemar tests with Bonferroni correction performed for all model comparisons.

TRIPOD Compliance Summary

Section	Items	Compliant	Partial/Action Required
Title & Abstract	1	1/1	—
Introduction	2	2/2	—
Methods	3–11	9/9	—
Results	12–16	5/5	—
Discussion	17	1/1	—
Other	18–19	1/2	Item 18 (Funding statement to be added)
Total	22	21/22	1 item to add (funding statement)

Overall TRIPOD Compliance: 21/22 items (95%) — 1 minor addition required

Reference: Collins, G. S., Reitsma, J. B., Altman, D. G., & Moons, K. G. (2015). Transparent reporting of a multivariable prediction model for individual prognosis or diagnosis (TRIPOD): The TRIPOD statement. BMJ, 350, g7594. <https://doi.org/10.1136/bmj.g7594>

Moons, K. G., Altman, D. G., Reitsma, J. B., Ioannidis, J. P., Macaskill, P., Steyerberg, E. W., ... & Collins, G. S. (2015). Transparent Reporting of a multivariable prediction model for Individual Prognosis or Diagnosis (TRIPOD): Explanation and elaboration. Annals of Internal Medicine, 162(1), W1–W73. <https://doi.org/10.7326/M14-0698>